

Samyak Sarnayak

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Professional Experience

Software Engineer, e6data Feb 2025 – Present

- Implemented Variant data type for our custom **query engine**, making it the first after Spark/Databricks to support querying **Open Variant** tables.
- Implemented **distributed join algorithms** - broadcast hash join and shuffled hash join.
- Implemented **dynamic switching of aggregates and joins** from non-partitioned to partitioned, improving perf by **~2x** in ClickBench and over **30%** in TPC-DS.
- Working on a **Rust** rewrite of the query engine - using Apache DataFusion, focusing on distributed execution.
- Authored multiple blog posts and gave a talk at Rust India Conference 2026.

Software Engineer, Normalyze Nov 2023 – Jan 2025

- Designed an incremental SaaS scanning system, handling 10k changes per hour.
- Built a data pipeline using Snowflake and dbt, processing **10M+ results** in 2 minutes.

Software Engineer, Chaos Genius Jul 2022 – Oct 2023

- Built a rule-based Snowflake warehouse optimization engine, saving **30%** in costs for several **Fortune 500s**.
- Provided cost-optimization recommendations for Snowflake **SQL queries**.
- Managed 10+ releases of both open source and proprietary products.

Education

B.Tech. in Computer Science and Engineering, PES University, 9.62 CGPA 2018 – 2022 | Bengaluru, India

Publications

Analysis of Garbage Collection Patterns to Extend Microbenchmarks for Big Data Workloads, ACM ICPE

- Presented at WOSP-C workshop held at ICPE (International Conference on Performance Engineering) 2022.
- Extended a custom JVM and a profiling tool, AntTracks, to provide per-object statistics.

Projects

otter-sql, SeaQL Jun 2022 – May 2023
Embeddable SQL executor written in Rust. Uses a custom intermediate code and instruction set.

michie, Over 10k downloads on crates.io Mar 2022 – Jun 2022
A Rust attribute macro that adds memoization to a function.

toipe, Over 600 stars on GitHub Jan 2022 – Present
A typing test terminal app written in Rust.

par-a-graph, PES University Mar 2021 – Jun 2021
Parallelized implementation of Pagerank algorithm in Go, achieving **4.2x** speedup on CPU and ~7x on GPU.

browser-history, Over 100k downloads on PyPI Aug 2020 – Present
Open Source Python Package for retrieving any browser's history and bookmarks.

Open Source

Apache DataFusion

Contributions to datafusion and the surrounding ecosystem like arrow-rs, sqlparser-rs, variant.

Meilisearch

Some fixes to search engine library - milli.